

Jurnal Praktikum

Sistem Operasi

Modul 5: Eksplorasi Proses Xinu

Tujuan:

1. Mahasiswa memahami Process Control Block (PCB) pada Xinu.
2. Mahasiswa mampu melakukan modifikasi sederhana pada process.

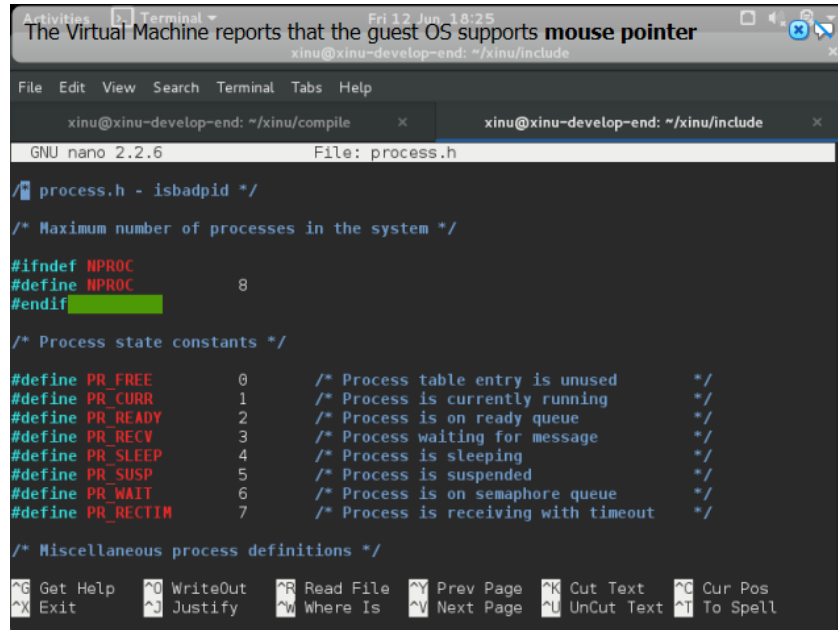
Catatan:

1. Praktikan wajib untuk screenshot setiap langkah yang dikerjakan hingga tampilan output akhir.
2. Untuk soal source code, kumpulkan SS-nya saja.
3. Praktikan wajib untuk melakukan screenshot lengkap dengan nama root. Contoh: root@username.
4. Berikan identitas nama - nim dalam bentuk comment di Source Code.
5. Harap kerjakan secara mandiri, jika tidak paham silahkan bertanya kepada Asisten Praktikum masing-masing. Dilarang mengcopy jawaban dan source code dari teman!

Jurnal

1. [10 Poin] Jawablah pertanyaan berikut ini:

- a. Berapa banyaknya maksimum proses yang ada pada Xinu?



```
GNU nano 2.2.6 File: process.h

/* process.h - isbadpid */

/* Maximum number of processes in the system */

#ifndef NPROC
#define NPROC 8
#endif

/* Process state constants */

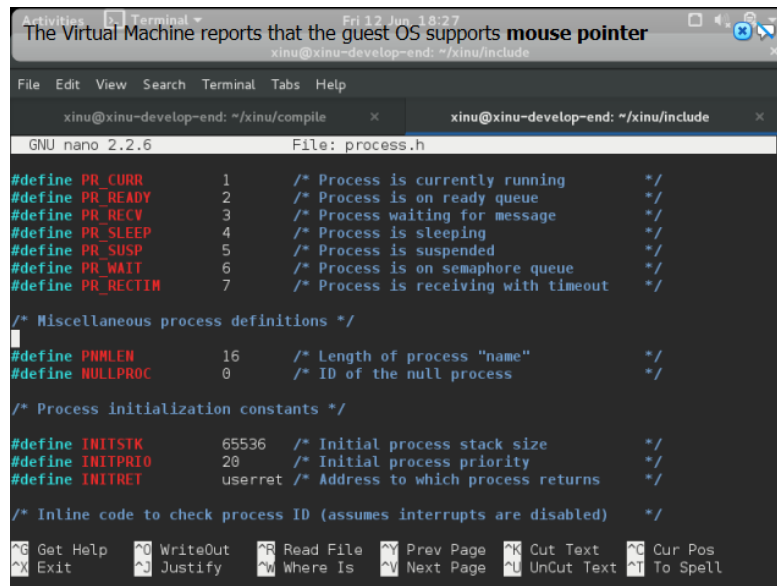
#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

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^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

Terdapat 8 maksimal proses

- b. Berapa maksimal panjang nama suatu proses pada Xinu?



```
GNU nano 2.2.6 File: process.h

#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

#define PNMLEN 16 /* Length of process "name" */
#define NULLPROC 0 /* ID of the null process */

/* Process initialization constants */

#define INITSTK 65536 /* Initial process stack size */
#define INITPRIO 20 /* Initial process priority */
#define INITRET userret /* Address to which process returns */

/* Inline code to check process ID (assumes interrupts are disabled) */

^G Get Help ^O WriteOut ^R Read File ^Y Prev Page ^K Cut Text ^C Cur Pos
^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell
```

Panjang nama proses memiliki batas maksimal berupa 16 huruf

- c. Berapa nilai prioritas awal pada saat proses dibuat?

```

GNU nano 2.2.6 File: process.h

#define NULLPROC 0 /* ID of the null process */

/* Process initialization constants */

#define INITSTK 65536 /* Initial process stack size */
#define INITPRIO 20 /* Initial process priority */
#define INITRET userret /* Address to which process returns */

/* Inline code to check process ID (assumes interrupts are disabled) */
#define isbadpid(x) ( ((pid32)(x) < 0) || \
                      ((pid32)(x) >= NPROC) || \
                      (proctab[(x)].prstate == PR_FREE))

/* Number of device descriptors a process can have open */
#define NDESC 5 /* Must be odd to make procent 4N bytes */

/* Definition of the process table (multiple of 32 bits) */

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```

Prioritas awal Ketika sebuah proses berjalan Adalah 20.

d. Ada berapa total state pada Xinu? Sebutkan!

```

development-system [Running] - Oracle VM VirtualBox : 1
GNU nano 2.2.6 File: process.h

/* process.h - isbadpid */

/* Maximum number of processes in the system */
#ifndef NPROC
#define NPROC 8
#endif

/* Process state constants */
#define PR_FREE 0 /* Process table entry is unused */
#define PR_CURR 1 /* Process is currently running */
#define PR_READY 2 /* Process is on ready queue */
#define PR_RECV 3 /* Process waiting for message */
#define PR_SLEEP 4 /* Process is sleeping */
#define PR_SUSP 5 /* Process is suspended */
#define PR_WAIT 6 /* Process is on semaphore queue */
#define PR_RECTIM 7 /* Process is receiving with timeout */

/* Miscellaneous process definitions */

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^X Exit ^J Justify ^W Where Is ^V Next Page ^U UnCut Text ^T To Spell

```

Delapan state, Namanya tertera pada screenshoot mulai dari free hingga rectim

2. [20 Poin] Perintah `ps` adalah perintah untuk menampilkan statistik process yang berjalan. Source code dari `ps` tersimpan pada file `xsh_ps.c`. Carilah file tersebut dan beri komentar pada 20 baris terakhir di source code tersebut!

```

printf("%3s %-16s %5s %4s %10s %-10s %10s\n",
      "-----", "-----", "-----", "-----",
      "-----", "-----", "-----");

/* Output information for each process */
for (i = 0; i < NPROC; i++) {
    prptr = 6proctab[i];
    if (prptr->prstate == PR_FREE) { /* Skip unused slots */
        continue;
    }
    printf("%3d %-16s %s %4d %4d 0x%08X 0x%08X %8d\n",
          i, prptr->prname, pstate[(int)prptr->prstate],
          prptr->prprio, prptr->prparent, prptr->prstkbase,
          prptr->prstkptr, prptr->prstklen);
}
return 0;
}

```

3. [35 Poin] Ubahlah perintah `ps` (source code: `xsh_ps.c`) pada Xinu sehingga menampilkan informasi tambahan berupa kolom yang berisi total message yang ada pada proses seperti gambar di bawah ini:

xsh \$ ps												
Pid	Name	State	Prio	Ppid	Stack	Base	Stack	Ptr	Stack	Size	Msg	Content
0	prnull	ready	0	0	0x005FDFFC	0x00FFFF04	8192	1	4			
1	rdsproc	susp	200	0	0x005FBFFC	0x005FBFC8	16384	0	0			
2	ipout	wait	500	0	0x005F7FFC	0x005F7F40	8192	0	0			
3	netin	wait	500	0	0x005F5FFC	0x005F5EE0	8192	0	0			
5	Main process	recv	20	4	0x005E3FFC	0x005E3F64	65536	0	0			
6	shell	recv	50	5	0x005D3FFC	0x005D3C7C	8192	0	0			
7	ps	curr	20	6	0x005F3FFC	0x005F3DC8	8192	0	0			

Kolom `Msg` adalah banyaknya pesan yang ada dalam proses.

Kolom `Content` adalah isi dari pesan tersebut.

Langkah pengerjaan:

- Modifikasi source code pada file `xsh_ps.c`
- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah `$ps`. Pastikan hasilnya sesuai dengan contoh output pada gambar yang diberikan.

- Screenshot source kode dan output akhir hasil modifikasi

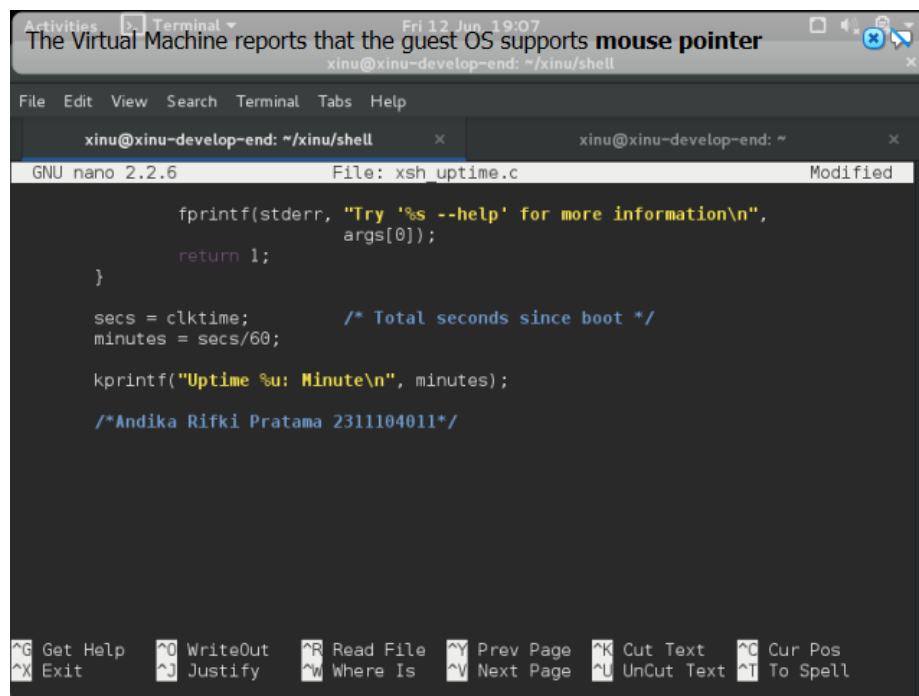
```
xsh $ ps
sps

==== ps modif aktif ====
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size  msg  content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait  500  0 0x005F9FFC 0x005F9ED0 8192
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536
5 shell           recv  50   4 0x005F7FFC 0x005F79AC 8192
6 ps              curr  20   5 0x005F5FFC 0x005F5FC4 8192
xsh $ command sps not found
```

4. [35 Poin] Ubahlah perintah uptime pada Xinu sehingga menampilkan lamanya Xinu sejak booting **hanya dalam satuan menit**.

Langkah pengerjaan:

- Modifikasi source code pada file xsh_uptime.c
- Kompilasi ulang Xinu dengan perintah seperti pada modul sebelumnya
- Jalankan Backend VM
- Setelah sistem berjalan, jalankan perintah `$uptime`. Pastikan hasilnya sesuai dengan contoh output yang diinginkan
- Screenshot source code dan output akhir hasil modifikasi



```
Activities | Terminal | Fri 12 Jun 19:07
The Virtual Machine reports that the guest OS supports mouse pointer
xinu@xinu-develop-end: ~/xinu/shell

File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/xinu/shell | xinu@xinu-develop-end: ~
GNU nano 2.2.6 | File: xsh_uptime.c | Modified

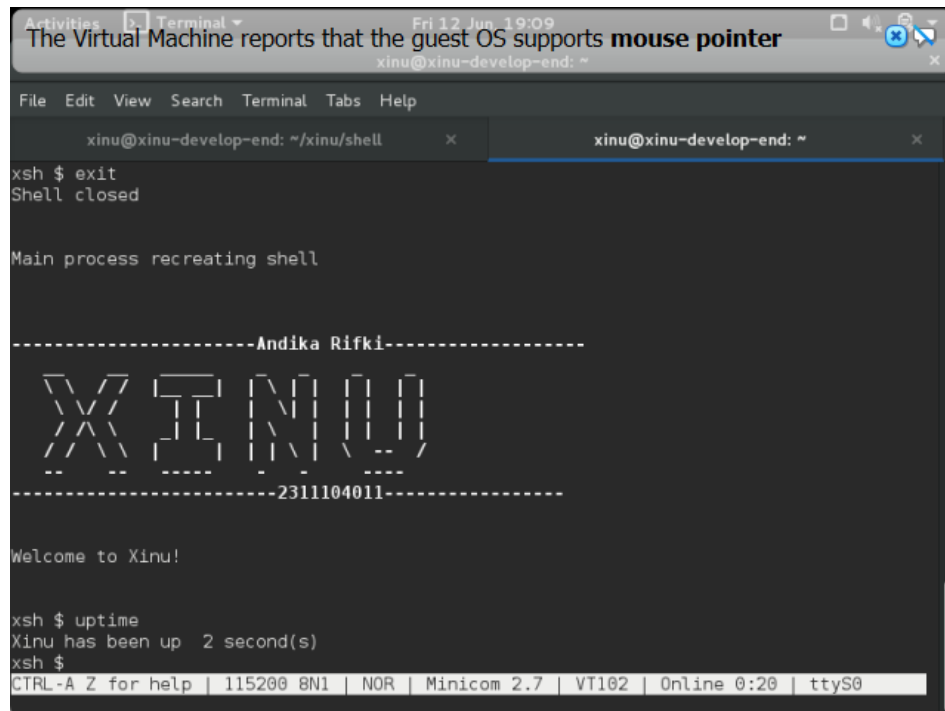
    fprintf(stderr, "Try '%s --help' for more information\n",
               args[0]);
    return 1;
}

secs = clktime;      /* Total seconds since boot */
minutes = secs/60;

kprintf("Uptime %u: Minute\n", minutes);

/*Andika Rifki Pratama 2311104011*/

^G Get Help  ^O WriteOut  ^R Read File  ^Y Prev Page  ^K Cut Text   ^C Cur Pos
^X Exit      ^J Justify   ^W Where Is  ^V Next Page  ^L UnCut Text ^T To Spell
```



```
Activities Terminal Fri 12 Jun 19:09
The Virtual Machine reports that the guest OS supports mouse pointer
xinu@xinu-develop-end: ~

File Edit View Search Terminal Tabs Help

xinu@xinu-develop-end: ~/xinu/shell x xinu@xinu-develop-end: ~ x
xsh $ exit
Shell closed

Main process recreating shell

-----Andika Rifki-----
XINU
-----2311104011-----

Welcome to Xinu!

xsh $ uptime
Xinu has been up 2 second(s)
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:20 | ttyS0
```